10TH CLASS COMPUTER SCIENCE — NEW SYLLABUS (PECTAA)

Chapter 6 — Introduction to AI and ML

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Multiple Choice Questions

1. What is Artificial Intelligence (AI)?

(a) Solving math problems by hand
(b) Machines performing tasks like humans
(c) Sending emails using a computer
(d) Watching videos online

Answer: (b) — Artificial Intelligence means the ability of the machines to perform the smart tasks which normally need human thinking, such as understanding language, recognising pictures and taking decisions. Sending an email or watching a video is only the normal use of a computer, not AI.

2. Which of the following is an example of AI?

(a) A chatbot replying to your questions
(b) Using a whiteboard
(c) A TV remote
(d) Typing on a keyboard

Answer: (a) — A chatbot is an AI program which understands the questions of the user and answers them by itself. A whiteboard, a TV remote and a keyboard are simple tools which only do what the user does with them.

3. What is Machine Learning (ML)?

(a) Teaching humans to use machines
(b) Teaching machines to learn from data like human
(c) Learning computer hardware basics
(d) A machine without internet

Answer: (b) — Machine Learning is a part of AI in which the machine is given data, and it learns from that data to make predictions or to find patterns, without being programmed for every single case.

4. Which app recommends videos using ML?

(a) Google Maps
(b) Microsoft Paint
(c) YouTube
(d) Windows Media Player

Answer: (c) — YouTube uses machine learning to study what you watch and to recommend the videos which you may like. Google Maps uses the traffic data for routes, MS Paint and Windows Media Player do not recommend anything.

studysapiens.com

5. In supervised learning, the data is:

(a) Unorganized
(b) Without any answers
(c) Labeled with correct answers
(d) Only text-based

Answer: (c) — In supervised learning the machine is given data along with the correct answers, called labels. The machine compares its own guesses with the correct answers and improves itself, as a student learns with a teacher.

6. Which of the following is an example of unsupervised learning?

(a) Predicting student marks
(b) Identifying spam emails
(c) Marking quizzes
(d) Grouping students by interest without labels

Answer: (d) — In unsupervised learning the data has no labels, and the machine groups the similar things by itself. Grouping the students by their interest without any labels is exactly this kind of work; predicting marks, finding spam and marking quizzes use the labelled data, so they are supervised.

7. Google Assistant is an example of:

(a) A game app
(b) A word processor
(c) A smart assistant using AI
(d) A calculator

Answer: (c) — Google Assistant is a smart assistant which uses AI to understand voice commands, translate languages, set reminders and answer questions.

8. AI is mostly used for:

(a) Handwriting notes
(b) Making phone calls
(c) Charging phones
(d) Smart decision making and automation

Answer: (d) — The main use of AI is smart decision making and automation, because AI systems study the data, find the patterns and take the decisions automatically, for example recommending products, detecting fraud or predicting diseases.

9. A chatbot on JazzCash is an example of:

(a) Cloud computing
(b) Data visualization
(c) AI in business
(d) Unsupervised learning

Answer: (c) — Many Pakistani banks, telecom companies and online stores use chatbots which answer the questions of the customers by themselves. This is a clear example of using AI in business to give quick customer service.

studysapiens.com

B. Short Answer Questions

1. What is the difference between AI and ML?

Artificial Intelligence (AI)	Machine Learning (ML)
AI is the wide field in which machines are made to perform the tasks which normally need human thinking.	ML is a part of AI in which the machine learns from the data.
It includes many techniques such as language understanding, vision and robotics.	It includes the algorithms such as linear regression, decision trees and K-means.
Its aim is to make the machine intelligent.	Its aim is to make the machine learn from the data and make predictions or find patterns without being programmed for every case.
Example: a robot which walks and talks.	Example: a model which predicts whether a student will pass, after learning from the old results.

2. Give one real-life example of AI used in Pakistan.

A very common example is the **chatbot on JazzCash**. When a customer writes a question about his balance, a transaction or a payment, the chatbot understands his question and answers it at once, without any human agent. In the same way, Pakistani banks and telecom companies use chatbots, and online shopping sites use AI to suggest the products which the customers may like.

3. How does supervised learning work?

In supervised learning the machine is trained with data which already contains the correct answers, called labels. The machine looks at the input, makes its own guess, and then compares the guess with the correct answer given in the data. The difference between the two is used to correct the model, and this process is repeated many times, so the accuracy of the model improves. When the training is complete, the model can predict the answer for a new input which it has never seen; for example, a model trained with the marks and attendance of the students can predict who might fail.

4. Define confusion matrix.

A confusion matrix is a simple table which shows how many predictions of a model were correct and how many were incorrect. It has four values: **TP (True Positive)**, when the model predicted pass and the student actually passed; **TN (True Negative)**, when it predicted fail and the student actually failed; **FP (False Positive)**, when it predicted pass but the student failed; and **FN (False Negative)**, when it predicted fail but the student passed. It tells us where the model is making mistakes, and it is used to calculate the accuracy of the model.

5. How can businesses use AI?

(i) Customer analysis: companies study the buying habits of the customers to find which products are sold more, so that they can keep the stock ready in the right city at the right time.

(ii) Chatbots and automated help: banks, telecom companies and online stores use chatbots which answer the questions of the customers at once and give them help without any delay.

(iii) Recommendations: online stores use machine learning to suggest the items which a customer often buys or may like, which increases the sales.

(iv) Better decisions: AI predicts the demand, detects the fraud and helps in planning, which saves the time and the cost of the business.

6. Give one example each of a smart assistant and a recommendation system.

Smart assistant: Google Assistant (or Siri) — it uses AI to understand the voice commands, to answer the questions, to translate the languages and to set the reminders.

Recommendation system: YouTube — it studies the videos you watch and suggests the other videos which you may like; in the same way, online stores suggest the products according to your previous purchases.

7. What kind of data is used in unsupervised learning?

In unsupervised learning the data is **unlabelled**, that is, it has no correct answers or labels with it. The machine is given only the raw data, such as the age and the interests of many students, and it has to find the patterns and group the similar things by itself. For example, it can group the students who have similar interests, without anyone telling it which group each student belongs to.

8. Define linear regression.

Linear regression is a machine learning algorithm which predicts a value on the basis of the straight-line relationship between two things. One thing is taken as the input (X) and the other as the output (Y). For example, a teacher can predict the marks of the students from the number of hours they studied: the study hours are the input X and the test score is the output Y. The algorithm finds the line which best fits the old data and then uses it to predict the marks of the students who studied, say, 6 hours.

C. Long Answer Questions

1. Explain supervised and unsupervised learning with suitable examples.

(a) Supervised learning: in supervised learning the machine is given the data along with the correct answers, which are called labels. The machine learns by comparing its own guesses with the correct answers, and it corrects itself again and again, so it improves over the time. It is like learning with a teacher, because the teacher tells the correct answer after every try.

Examples: a model trained with the pictures of apples and mangoes can later tell whether a new picture is of an apple or a mango. In the same way, a system trained

with the marks and the attendance of the old students can predict which student might fail, and a filter trained with the labelled emails can identify a spam email.

(b) Unsupervised learning: in unsupervised learning the machine is given only the data and there are no labels with it. The machine has to find the patterns and group the similar things (clusters) by itself, without being told what the correct answer is. It is like learning on your own, by exploring.

Examples: a website can group its customers according to their buying habits without any labels, so it can target each group separately. In the same way, the students can be grouped according to their interests, and the K-means algorithm is used for such clustering.

Basis	Supervised learning	Unsupervised learning
Data	Labelled data, with the correct answers	Unlabelled data, without any answers
Aim	To predict the correct output for a new input	To find the hidden patterns and group similar items
Guidance	Works like learning with a teacher	Works like learning on your own
Algorithms	Linear regression, decision trees	K-means clustering
Example	Predicting the marks or the spam emails	Grouping the students by their interests

2. Explain how model performance is evaluated in machine learning using the following metrics: (a) Confusion Matrix, (b) Accuracy.

After the training of a machine learning model, it is necessary to measure how accurate and reliable it is. For this purpose the following metrics are used.

(a) Confusion matrix: a confusion matrix is a simple table which shows how many predictions of the model were correct and how many were wrong. For a problem in which a student has to be classified as pass or fail, the table has the following four parts:

Value	Meaning
TP (True Positive)	The model predicted **pass**, and the student actually passed.
TN (True Negative)	The model predicted **fail**, and the student actually failed.
FP (False Positive)	The model predicted **pass**, but the student actually failed.
FN (False Negative)	The model predicted **fail**, but the student actually passed.

The TP and TN are the correct predictions, while the FP and FN are the mistakes. In this way the confusion matrix tells us not only how many mistakes were made but also which kind of mistake was made, which helps in improving the model.

(b) Accuracy: accuracy tells us how many predictions were correct out of all the predictions made by the model. It is calculated by the following formula:

$$\text{Accuracy (A)} = (TP + TN) \div (TP + TN + FP + FN)$$

For example, if out of 20 students the model gives TP = 8, TN = 6, FP = 4 and FN = 2, then A = (8 + 6) ÷ 20 = 14 ÷ 20 = 0.70, that is **70 %**. A higher accuracy means that the model is more reliable, and such a model can be used for making the decisions.

3. A model predicts if students will pass (1) or fail (0). Out of 20 students: True Positives (TP) = 8, True Negatives (TN) = 6, False Positives (FP) = 4, False Negatives (FN) = 2. You are required to (a) fill the confusion matrix, and (b) calculate accuracy.

(a) Confusion matrix:

	Actual: Pass (1)	**Actual: Fail (0)**
Predicted: Pass (1)	TP = 8	FP = 4
Predicted: Fail (0)	FN = 2	TN = 6

Here the total number of students is 8 + 6 + 4 + 2 = 20, which matches the number given in the question, so the matrix is filled correctly.

(b) Accuracy:

$$\text{Accuracy} = (TP + TN) \div (TP + TN + FP + FN)$$
$$= (8 + 6) \div (8 + 6 + 4 + 2)$$
$$= 14 \div 20$$
$$= 0.70 \;=\; \mathbf{70\ \%}$$

So the model made 14 correct predictions out of 20, and its accuracy is 70 %. The model still made 6 mistakes, that is 4 false positives and 2 false negatives, so it can be improved further by training it with more data.